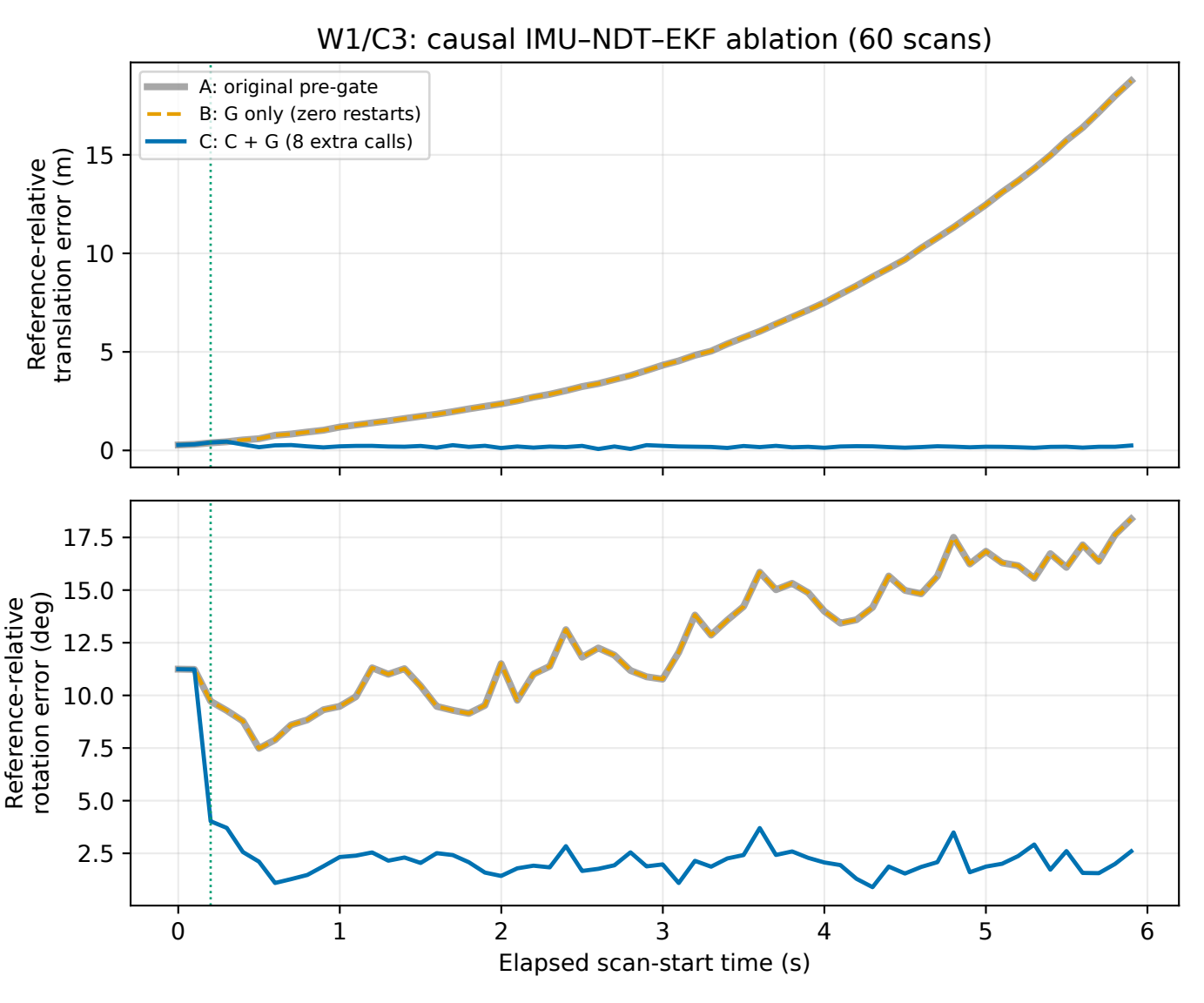
W1/C3: causal IMU–NDT–EKF ablation (60 scans)
A: original pre-gate
B: G only (zero restarts)
C: C + G (8 extra calls)
Reference-relative translation error (m)
0
5
10
15
Reference-relative rotation error (deg)
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
1
2
3
4
5
6
Elapsed scan-start time (s)